

PRODUCT SENTIMENT ANALYSIS AND FEATURE ROADMAP PRIORITIZATION

Using Python | Case Study: FitLife Health & Fitness Applications

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Presentation Roadmap

- Background, problem statement and objectives
- Data source and data preparation
- Methodology and system workflow
- Implementation: VADER, TF-IDF and RICE
- Results and output screenshots
- Testing, managerial implications and conclusion
- Future scope and key takeaways

1. Introduction & Problem Statement

- Health & Fitness applications receive large volumes of unstructured user reviews.
- Manual review reading makes it difficult to identify recurring issues and opportunities at scale.
- The project converts review text into structured sentiment, feature-related terms and a prioritized roadmap.
- The focus is on evidence-based product improvement using Python and NLP.

272

final reviews analyzed

8

matched applications

70.6%

positive sentiment

147.33

avg. review length



2. Objectives

- Clean and prepare Google Play Store app and user-review data.
- Classify reviews into Positive, Negative and Neutral sentiment using VADER.
- Identify important recurring terms using TF-IDF feature extraction.
- Prioritize the extracted terms using the RICE framework.
- Translate analytical results into product and managerial insights.

3. Data Source & Dataset

- Source: Google Play Store Apps dataset hosted on Kaggle.
- Application file: googleplaystore.csv — 10,841 records and 13 columns.
- Review file: googleplaystore_user_reviews.csv — 64,295 records and 5 columns.
- After review cleaning: 27,307 usable reviews.
- Health & Fitness filtering produced 254 applications; all had at least 100 reviews.
- Top 30 applications were selected by review count; exact matching yielded 8 applications and 272 final reviews.

Kaggle: Google Play Store Apps — <https://www.kaggle.com/datasets/lava18/google-play-store-apps>

4. Proposed Methodology & System Design



- Python libraries: Pandas, NumPy, NLTK/VADER, Scikit-learn, Matplotlib and Seaborn.
- The workflow is reproducible and moves from raw review text to product prioritization.

5. Implementation & Key Python Logic

```
# Load data
apps = pd.read_csv('/content/googleplaystore.csv')
reviews = pd.read_csv('/content/googleplaystore_user_reviews.csv')

# Health & Fitness + Top 30
health = apps[apps['Category'] == 'HEALTH_AND_FITNESS']
top30 = health[health['Reviews'] >= 100].sort_values(
    'Reviews', ascending=False).head(30)

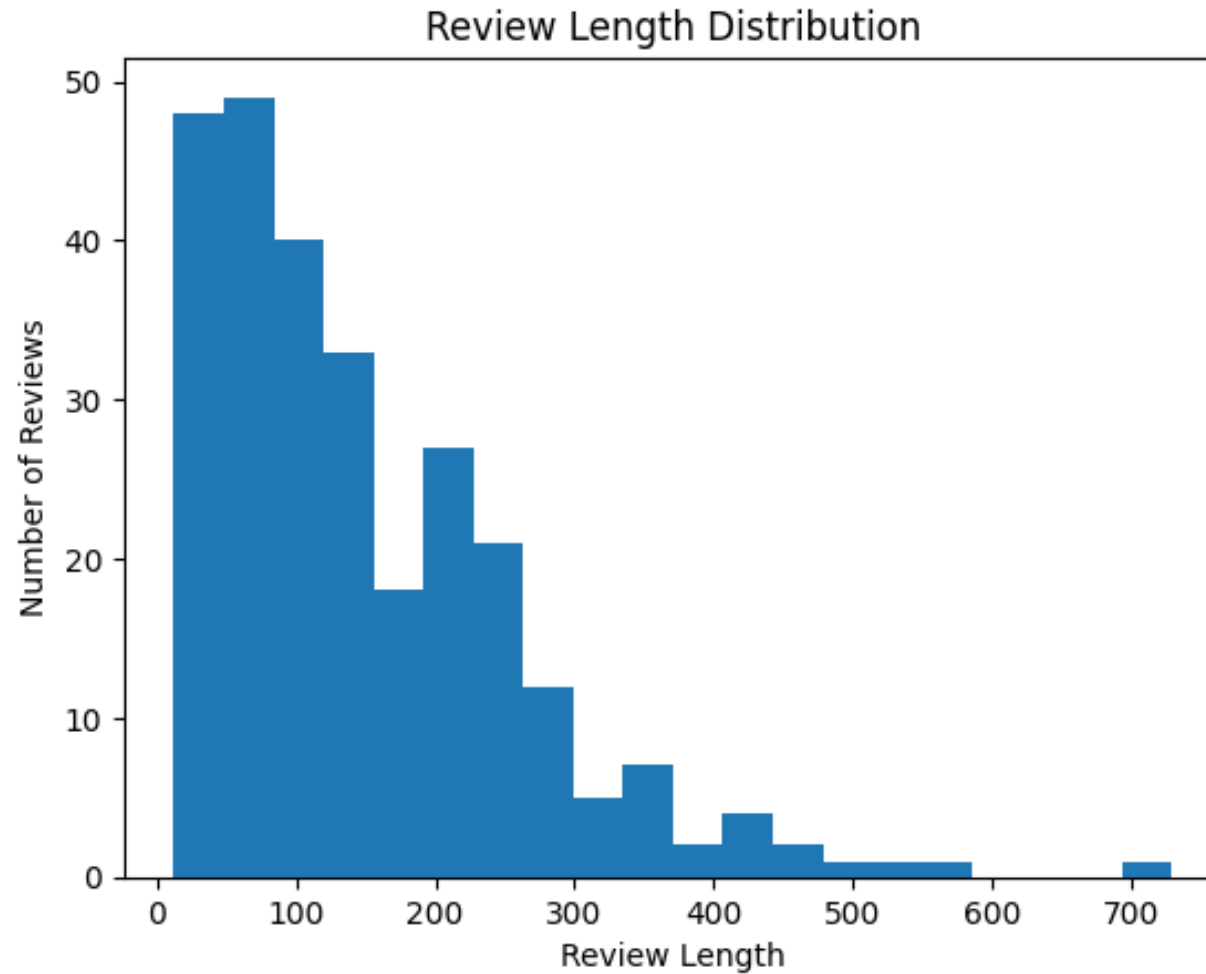
# Match reviews
final = reviews[reviews['App'].isin(top30['App'])].copy()

# VADER
final['VADER_Score'] = final['Translated_Review'].apply(
    lambda x: analyzer.polarity_scores(x)['compound']
)
```

VADER thresholds: Positive ≥ 0.05 | Negative ≤ -0.05 | Otherwise Neutral

6. Results: Review Length

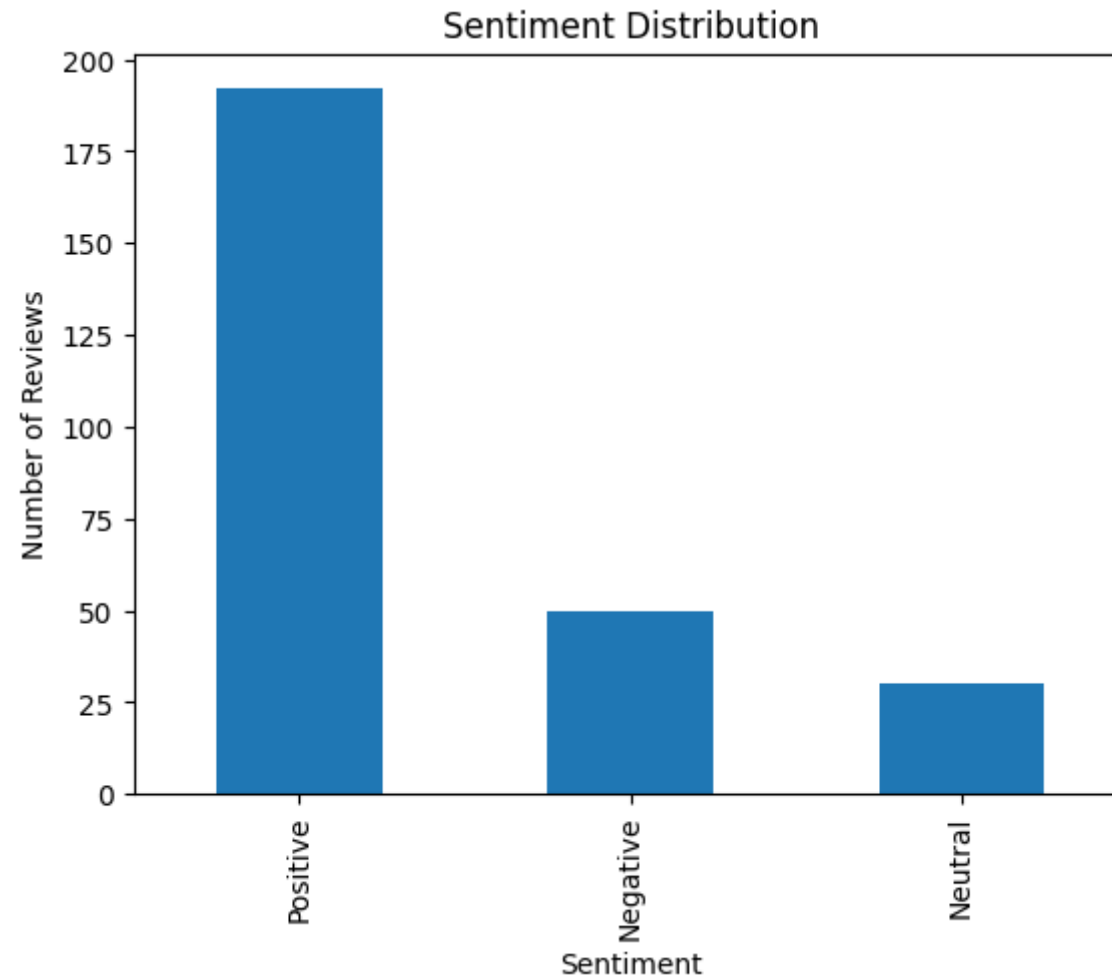
Distribution of review text length in the final dataset



User-provided Python output. Average review length = 147.33 characters.

7. Results: VADER Sentiment

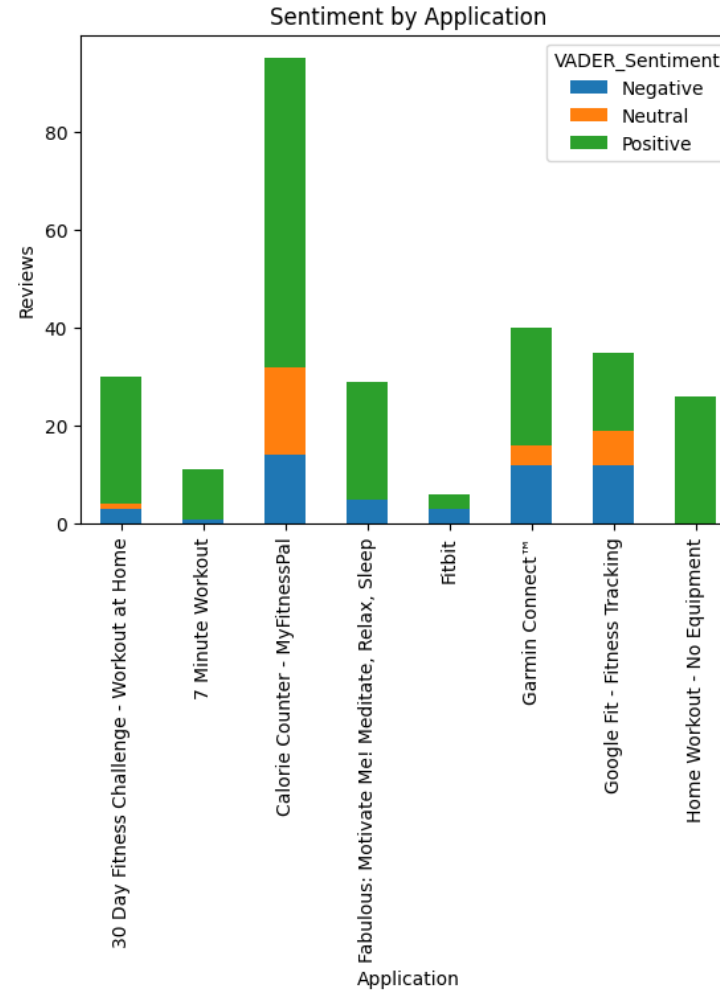
Sentiment classification of the 272 analyzed reviews



User-provided Python output. Positive = 192, Negative = 50, Neutral = 30.

8. Results: Sentiment by Application

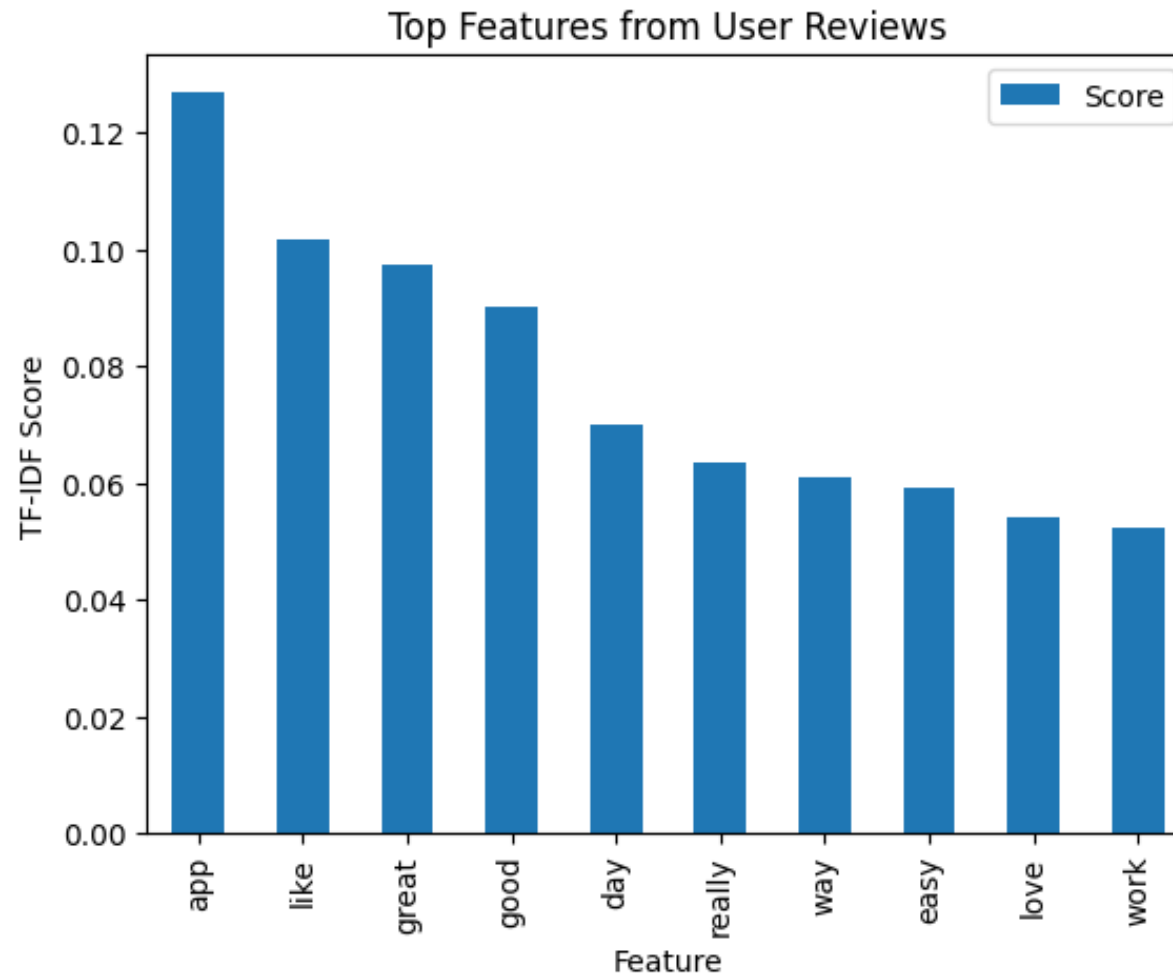
Comparison of sentiment patterns across the matched applications



User-provided Python output for application-level sentiment comparison.

9. Results: TF-IDF Feature Extraction

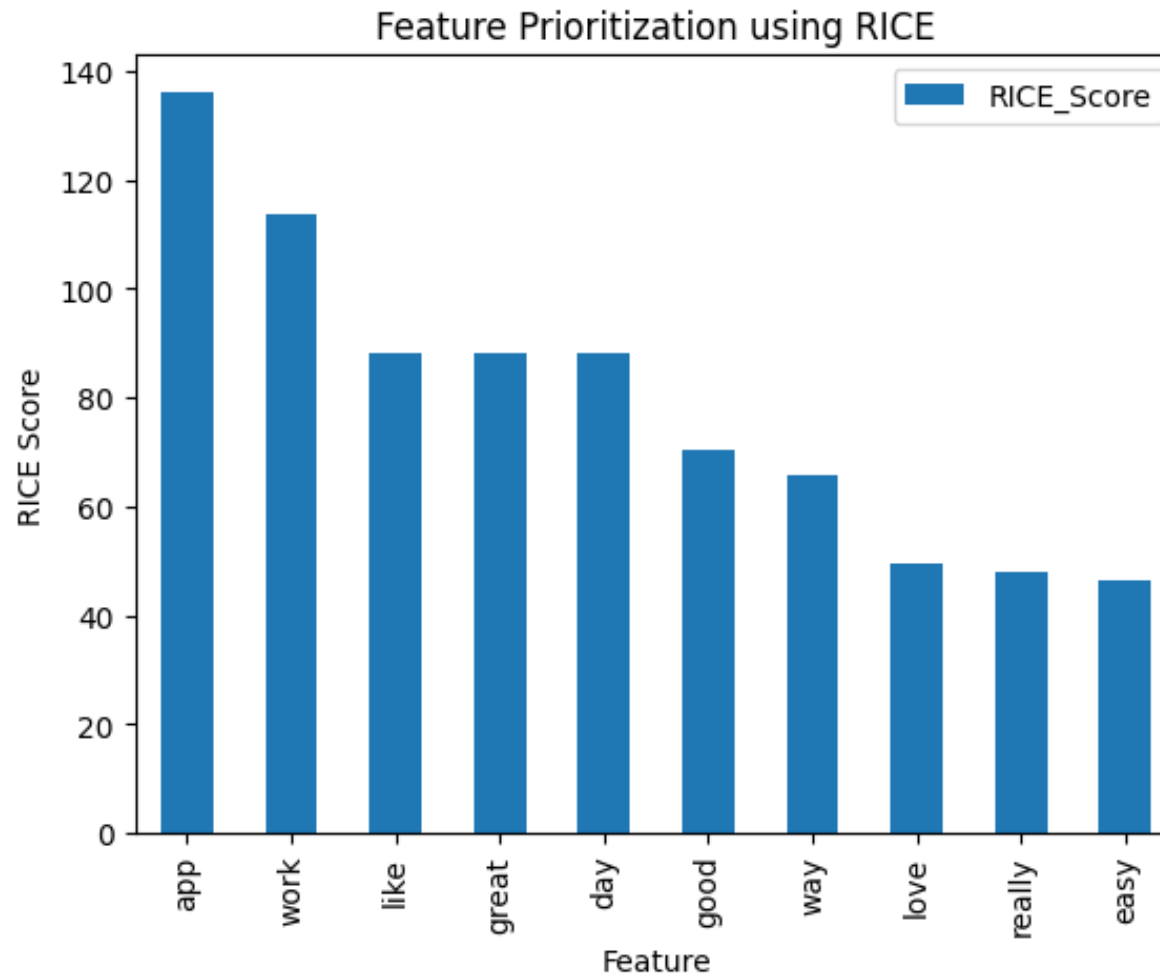
Top terms identified from user reviews



User-provided Python output. TF-IDF was used to surface recurring important terms.

10. Results: RICE Prioritization

Prioritizing the top extracted terms for product discussion



User-provided Python output. $RICE = Reach \times Impact \times Confidence \div Effort$.

11. Key Results & Interpretation

- Final analytical dataset: 272 reviews across 8 matched Health & Fitness applications.
- Sentiment: 192 Positive (70.6%), 50 Negative (18.4%), 30 Neutral (11.0%).
- Average review length: 147.33 characters.
- TF-IDF surfaced terms including app, like, great, good, day, really, way, easy, love and work.
- Top RICE score: app = 136.0; followed by work = 113.6 and like/great/day = 88.0 each.
- Because Impact, Confidence and Effort were held constant, the RICE ranking is primarily driven by Reach.



12. Managerial Implications

- Use positive reviews to understand which experiences create user value and engagement.
- Use negative reviews to identify usability, functionality and experience pain points.
- Recurring terms can guide deeper qualitative review and feature-level investigation.
- RICE provides a transparent structure for discussing which opportunities should be considered first.
- Product teams should validate keyword-based findings with domain knowledge and actual review context before implementation.

13. Testing & Validation

Test case	Outcome	Status
Dataset loading	10,841 apps / 64,295 reviews	Pass
Review cleaning	27,307 usable reviews	Pass
Health & Fitness filter	254 applications	Pass
Top-30 selection	30 applications	Pass
Review matching	8 applications	Pass
Final dataset	272 reviews	Pass
VADER scores	0 missing scores	Pass
Sentiment labels	0 missing labels	Pass
TF-IDF	20 features generated	Pass
RICE	10 ranked terms	Pass



14. Limitations

- The dataset may not represent current reviews or the full population of Health & Fitness app users.
- Exact app-name matching reduced the final review sample to 8 of the selected top 30 applications.
- VADER may struggle with sarcasm, ambiguity, mixed sentiment and domain-specific language.
- TF-IDF identifies terms, not complete product features; context must be reviewed separately.
- Constant RICE Impact, Confidence and Effort values make the current prioritization a simplified initial ranking.

15. Conclusion & Future Scope

- The project successfully converts unstructured app reviews into structured sentiment and feature-prioritization insights.
- The final analysis showed predominantly positive sentiment, while negative reviews provide opportunities for improvement.
- TF-IDF and RICE together provide a practical bridge from review text to product roadmap discussion.
- Future work: aspect-based sentiment analysis, transformer models, topic modeling, richer RICE estimates and larger/recent datasets.
- The workflow can be extended into a recurring product-feedback monitoring system.

16. References & Project Files

- Gupta, L. (2019). "Google Play Store Apps." Kaggle.
- Hutto, C. J., & Gilbert, E. E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text.
- Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval.
- Python, Pandas, Scikit-learn and VADER documentation.

Python + Excel/Data Files

Google Drive project folder: https://drive.google.com/drive/folders/1CtfboqwPOq2IKGxUG93sk82Ex1kxanrP?usp=drive_link

The folder contains the project source files and downloaded data files.